

Amity University Punjab, Mohali

B.Sc. Mathematics (5th Semester)
Set Theory and Metric Spaces (MAT304)

Unit III: Complete Metric Spaces

and Continuous Functions

Dr. Lokesh Singh

lsingh@pb.amity.edu

October 28, 2026

Contents

1 Sequences in Metric Spaces **2**
- 1.1 Motivation: Why Sequences? . . . 2
- 1.2 Convergent Sequences . . . 2
- 1.3 Cauchy Sequences . . . 3
- 1.4 The Relationship Between Convergence and Cauchy . . . 3

2 Complete Metric Spaces **4**
- 2.1 Definition and Motivation . . . 4
- 2.2 Examples of Complete and Incomplete Spaces . . . 5
- 2.3 Closed Subspaces of Complete Spaces . . . 5
- 2.4 Cantor's Intersection Theorem . . . 6

3 Dense Sets and Separable Spaces **7**
- 3.1 Dense Subsets . . . 7
- 3.2 Separable Spaces . . . 7

4 Nowhere Dense Sets and Baire's Category Theorem **8**
- 4.1 Nowhere Dense Sets . . . 8
- 4.2 Meager and Non-Meager Sets . . . 8
- 4.3 Baire's Category Theorem . . . 9
- 4.4 Applications of Baire Category . . . 9

5 Continuous and Uniformly Continuous Functions **10**
- 5.1 Continuity — Review . . . 10
- 5.2 Uniform Continuity . . . 10
- 5.3 Heine–Cantor Theorem . . . 11
- 5.4 Preservation of Cauchy Sequences . . . 11

6 Homeomorphisms **12**
- 6.1 Definition and Intuition . . . 12
- 6.2 Examples and Non-Examples . . . 12
- 6.3 Topological Properties (Invariants) . . . 13

7 The Banach Contraction Principle **13**
- 7.1 The Fixed-Point Idea . . . 13
- 7.2 Statement and Proof . . . 13
- 7.3 Examples and Applications . . . 15

8 Exercises **15**

1 Sequences in Metric Spaces

(Lectures 1–2)

1.1 Motivation: Why Sequences?

In Unit II, we learned that a set is open iff every point has a small ball around it contained in the set. But how do we **detect** boundaries, limits, or the behaviour of a space? Sequences provide one of the most powerful tools for this.

- In $\mathbb{R}$, the statement "$x_n \to x$" captures the intuitive idea that the numbers "get arbitrarily close to x".
- In a general metric space, sequences test whether a point is a limit point, whether a set is closed, whether a function is continuous, and—most importantly—whether the space has any "gaps" (i.e., whether it is *complete*).

The whole of Unit III hinges on one simple question: **given a sequence (x_n) whose terms get arbitrarily close to each other, can we always guarantee that the sequence has a limit in the space?** The answer is *no* in general; the spaces where the answer is *yes* are precisely the **complete metric spaces**.

1.2 Convergent Sequences

Definition 1.1 (Convergent Sequence)**.** Let (X, d) be a metric space. A sequence $(x_n)_{n=1}^{\infty}$ in X **converges** to $x \in X$, written $x_n \to x$ or $\lim_{n\to\infty} x_n = x$, if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x) < \varepsilon \quad \text{for all } n \geq N.$$

The point x is called the **limit** of the sequence.

Remark 1.1. Equivalently: $x_n \to x$ iff $d(x_n, x) \to 0$ as a sequence of real numbers. This reduction to real numbers is often useful in computations.

Example 1.2 (Convergence in $\mathbb{R}$)**.** In $\mathbb{R}$ with the usual metric,

$$x_n = \frac{1}{n} \to 0, \quad y_n = 1 - \frac{1}{2^n} \to 1, \quad z_n = (-1)^n \text{ does } \textit{not} \text{ converge.}$$

Example 1.3 (Convergence in a Discrete Space)**.** In a discrete metric space, the only convergent sequences are those that are *eventually constant*. Indeed, if $x_n \to x$, then for $\varepsilon = 1/2$, we need $d(x_n, x) < 1/2$; in the discrete metric this forces $x_n = x$. So $x_n = x$ for all $n \geq N$.

Uniqueness of Limits. If $x_n \to x$ and $x_n \to y$ in a metric space, then $x = y$.

Proof. Suppose $x \neq y$. Let $\varepsilon = d(x, y)/2 > 0$. Since $x_n \to x$, there is N_1 with $d(x_n, x) < \varepsilon$ for $n \geq N_1$. Since $x_n \to y$, there is N_2 with $d(x_n, y) < \varepsilon$ for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$,

$$d(x, y) \leq d(x, x_n) + d(x_n, y) < \varepsilon + \varepsilon = 2\varepsilon = d(x, y),$$

a contradiction. Hence $x = y$. □

1.3 Cauchy Sequences

The definition of convergence **requires knowing the limit in advance**. But in most situations we only have the sequence itself—we do not know what point it will converge to. This motivates a purely *internal* condition.

Definition 1.2 (Cauchy Sequence)**.** A sequence (x_n) in a metric space is **Cauchy** if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_m, x_n) < \varepsilon \quad \text{for all } m, n \geq N.$$

Intuition. A Cauchy sequence captures the idea: *the terms eventually get arbitrarily close to each other.* Unlike convergence, we do **not** need to know the limit in advance.

Example 1.4 (Cauchy Sequence in $\mathbb{R}$ that Converges)**.** $x_n = 1/n$ is Cauchy and converges to 0.

Example 1.5 (Cauchy Sequence in $\mathbb{Q}$ that Does Not Converge in $\mathbb{Q}$)**.** Consider the sequence

$$x_1 = 1, \quad x_2 = 1.4, \quad x_3 = 1.41, \quad x_4 = 1.414, \quad \dots$$

of decimal truncations of $\sqrt{2}$. As a sequence of *real* numbers, it converges to $\sqrt{2}$. But $\sqrt{2} \notin \mathbb{Q}$, so the sequence does **not** converge in $\mathbb{Q}$. This shows that a Cauchy sequence in a metric space need not converge in that space.

1.4 The Relationship Between Convergence and Cauchy

Theorem 1.6 (Convergent implies Cauchy)**.** Every convergent sequence in a metric space is Cauchy.

Proof. Let $x_n \to x$. Given $\varepsilon > 0$, choose N such that $d(x_n, x) < \varepsilon/2$ for all $n \geq N$. Then for $m, n \geq N$,

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So (x_n) is Cauchy. □

The converse fails in general. A Cauchy sequence need not converge. Counterexample: the sequence $x_n = 1 + 1/2! + \cdots + 1/n!$ is Cauchy in $\mathbb{Q}$ but converges (in $\mathbb{R}$) to e, which is irrational—so no limit exists in $\mathbb{Q}$. This failure distinguishes complete metric spaces from incomplete ones.

Theorem 1.7 (Subsequences Inherit Limit and Cauchy Property)**.** (a) If $x_n \to x$, then every subsequence of (x_n) also converges to x.

(b) If (x_n) is Cauchy, then every subsequence of (x_n) is also Cauchy.

Proof. (a) Let (x_{n_k}) be a subsequence. Given $\varepsilon > 0$, choose N with $d(x_n, x) < \varepsilon$ for $n \geq N$. Since $n_k \to \infty$, there is K with $n_k \geq N$ for $k \geq K$. Then $d(x_{n_k}, x) < \varepsilon$ for $k \geq K$.

(b) The same N from the Cauchy definition works for any subsequence. □

Lemma 1.8 (Cauchy plus convergent subsequence implies convergence)**.** If a Cauchy sequence (x_n) has a subsequence (x_{n_k}) converging to x, then the whole sequence (x_n) converges to x.

Proof. Given $\varepsilon > 0$, choose N such that $d(x_m, x_n) < \varepsilon/2$ for $m, n \geq N$. Since $x_{n_k} \to x$, choose k with $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$. Then for all $n \geq N$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $x_n \to x$. □

Why is this lemma useful? It gives a tool to check convergence using Cauchy sequences: if we can prove (i) (x_n) is Cauchy and (ii) it has a convergent subsequence, then the whole sequence converges. This lemma is used in compactness arguments and in the proof of the Banach contraction principle.

Theorem 1.9 (Cauchy Sequences are Bounded)**.** Every Cauchy sequence in a metric space is bounded.

Proof. Apply the Cauchy definition with $\varepsilon = 1$. There exists N such that $d(x_m, x_N) < 1$ for all $m \geq N$. Let

$$R = \max\{d(x_1, x_N), \ldots, d(x_{N-1}, x_N), 1\} + 1.$$

Then $x_n \in B_R(x_N)$ for all n, so the sequence is bounded. □

2 Complete Metric Spaces

(Lectures 3–4)

2.1 Definition and Motivation

Definition 2.1 (Complete Metric Space)**.** A metric space (X, d) is **complete** if every Cauchy sequence in X converges to a point of X.

Intuition. Completeness is the metric-space analog of "having no holes." The space $\mathbb{R}$ is complete; the space $\mathbb{Q}$ is not—because there are Cauchy sequences in $\mathbb{Q}$ whose "limit would be $\sqrt{2}$," but $\sqrt{2} \notin \mathbb{Q}$. The completion of $\mathbb{Q}$ is precisely $\mathbb{R}$.

Why is completeness important?

- **Analysis.** The whole of real analysis (limits, derivatives, integrals, series) rests on the completeness of $\mathbb{R}$. Without it, the intermediate value theorem, extreme value theorem, and the fundamental theorem of calculus collapse.
- **Existence of solutions.** Completeness is the key hypothesis of the Banach contraction principle, which proves existence and uniqueness of solutions to differential equations, integral equations, and optimisation problems.
- **Functional analysis.** Banach spaces (complete normed vector spaces) and Hilbert spaces (complete inner-product spaces) are the central objects of mod-

ern analysis—and both require completeness.

2.2 Examples of Complete and Incomplete Spaces

Example 2.1 ($\mathbb{R}$ is Complete)**.** This is the fundamental example. Every Cauchy sequence of real numbers converges to a real number. (Proof relies on the supremum property of $\mathbb{R}$.)

Example 2.2 ($\mathbb{Q}$ is Not Complete)**.** The Cauchy sequence of decimal truncations of $\sqrt{2}$ does not converge in $\mathbb{Q}$.

Example 2.3 ($\mathbb{R}^n$ is Complete)**.** Every Cauchy sequence of vectors in $\mathbb{R}^n$ converges component-wise to a limit vector. This follows from $\mathbb{R}$'s completeness.

Example 2.4 (Discrete Metric Space is Complete)**.** Every Cauchy sequence in a discrete metric space must be eventually constant. Eventually constant sequences converge. Hence every discrete space is complete.

Example 2.5 ($\mathbb{Q}$ with the discrete metric is complete, but with the usual metric it is incomplete)**.** This shows that the same underlying set can be complete or incomplete depending on the metric. **Completeness is a metric-dependent property**.

Example 2.6 (Function Spaces)**.** Let $X = C([0,1])$ with the supremum metric $d_\infty(f,g) = \sup_{x\in[0,1]} |f(x)-g(x)|$. This space is **complete** (uniform limits of continuous functions are continuous). But X with the L^1 metric $d_1(f,g) = \int_0^1 |f(x)-g(x)|\,dx$ is **not complete**—a sequence of continuous functions can converge in L^1 to a discontinuous function.

Real-world: Image processing. CCTV footage is processed with denoising algorithms that work in function spaces. The completeness of $C([0,1])$ under the supremum norm guarantees that iterative image-enhancement procedures converge to a stable continuous image.

2.3 Closed Subspaces of Complete Spaces

Theorem 2.7 (Closed Subspace of Complete Space is Complete)**.** If X is a complete metric space and $F \subseteq X$ is closed, then F (with the induced metric) is complete.

Proof. Let (x_n) be Cauchy in F. It is also Cauchy in X, so by completeness of X, $x_n \to x$ for some $x \in X$. Since all $x_n \in F$ and F is closed, the limit $x \in F$. Hence (x_n) converges in F. □

Theorem 2.8 (Complete Subspace of Arbitrary Space is Closed)**.** If X is a metric space and $F \subseteq X$ is complete (with the induced metric), then F is closed in X.

Proof. Let $x \in \overline{F}$. Then there exists a sequence (x_n) in F with $x_n \to x$ in X. Since (x_n) is convergent in X, it is Cauchy in X, hence Cauchy in F. Since F is complete, (x_n) converges in F to some $y \in F$. But limits are unique in X, so $x = y \in F$. Hence F is closed. □

Combined result. For $F \subseteq X$ with X complete:

$$F \text{ is complete} \iff F \text{ is closed.}$$

2.4 Cantor's Intersection Theorem

Context. In $\mathbb{R}$, the familiar *nested intervals theorem* says that nested closed intervals with lengths shrinking to zero have exactly one common point:

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots, \quad b_n - a_n \to 0 \implies \bigcap_n [a_n, b_n] = \{\text{one point}\}.$$

Cantor's intersection theorem generalises this to complete metric spaces. It is one of the most useful tools for proving existence of points (e.g., in fractal geometry and in the proof of Baire's theorem).

Theorem 2.9 (Cantor's Intersection Theorem)**.** Let (X, d) be a complete metric space and (F_n) a decreasing sequence of non-empty closed subsets:

$$F_1 \supseteq F_2 \supseteq F_3 \supseteq \cdots.$$

If $\operatorname{diam}(F_n) \to 0$, then $\bigcap_{n=1}^{\infty} F_n$ consists of exactly one point.

Proof. **Step 1: Existence.** For each n, choose $x_n \in F_n$. We claim (x_n) is Cauchy. Given $\varepsilon > 0$, choose N such that $\operatorname{diam}(F_N) < \varepsilon$. For $m, n \geq N$, both $x_m, x_n \in F_N$ (since (F_n) is decreasing and $m, n \geq N$). Hence

$$d(x_m, x_n) \leq \operatorname{diam}(F_N) < \varepsilon.$$

So (x_n) is Cauchy.

Since X is complete, $x_n \to x$ for some $x \in X$. For each fixed k, all x_n with $n \geq k$ lie in F_k, and F_k is closed, so the limit $x \in F_k$. Hence $x \in \bigcap_k F_k$.

Step 2: Uniqueness. If $x, y \in \bigcap_k F_k$, then $x, y \in F_n$ for all n, so

$$d(x, y) \leq \operatorname{diam}(F_n) \to 0.$$

Hence $d(x, y) = 0$, i.e., $x = y$. □

What happens if we drop "$\operatorname{diam}(F_n) \to 0$"? The intersection can become empty. Example: in $\mathbb{R}$, $F_n = [n, \infty)$ is closed and decreasing, but $\bigcap F_n = \emptyset$. Hence the diameter condition is essential.

Application: Cantor set (revisited). Take $X = [0, 1]$ (complete) and $F_n = C_n$, the n-th stage of the Cantor set construction (a closed subset of $[0, 1]$). Then $\operatorname{diam}(C_n) = (1/3)^n \to 0$, and Cantor's intersection theorem gives that $\bigcap C_n = \mathcal{C}$ is non-empty and closed.

Application: Fractals via Iterated Function Systems. The Cantor set, Sierpinski triangle, and Koch snowflake are all limits of nested closed sets with diameters tending to zero—hence Cantor's intersection theorem guarantees they are non-empty compact sets.

3 Dense Sets and Separable Spaces

(Lecture 5)

3.1 Dense Subsets

Definition 3.1 (Dense Subset)**.** A subset $D \subseteq X$ is **dense** in X if $\overline{D} = X$. Equivalently, for every non-empty open set $U \subseteq X$, $U \cap D \neq \emptyset$. Equivalently, every point of X is a limit of some sequence in D.

Example 3.1 ($\mathbb{Q}$ is Dense in $\mathbb{R}$)**.** Every real number x is the limit of a sequence of rationals (its decimal truncations). Equivalently, every open interval (a, b) contains a rational.

Example 3.2 (Polynomials are Dense in $C([0, 1).$ $)]$ This is the **Weierstrass Approximation Theorem**: every continuous function on a closed interval can be uniformly approximated by polynomials.

Remark 3.3. The density of $\mathbb{Q}$ in $\mathbb{R}$ is what allows us to compute continuous functions via rational approximations—a key idea in numerical analysis. Every real-analysis algorithm that "samples at rational points" relies on density.

3.2 Separable Spaces

Definition 3.2 (Separable Space)**.** A metric space is **separable** if it has a *countable* dense subset.

Example 3.4. $\mathbb{R}^n$ is separable (dense subset: $\mathbb{Q}^n$).

Example 3.5. $C([0, 1])$ is separable (dense subset: polynomials with rational coefficients—countable by Unit I).

Example 3.6 (Not Separable)**.** ℓ^∞ (the space of bounded sequences with supremum metric) is not separable. Consider the uncountable set of sequences $\{e_A : A \subseteq \mathbb{N}\}$, where e_A is the characteristic function of A. For distinct A, B, $\|e_A - e_B\|_\infty = 1$. In a separable space, any countable dense set D would need at least one element in each ball $B_{1/3}(e_A)$—but such balls are pairwise disjoint, forcing an injection from the uncountable set $\{e_A\}$ into D, contradiction.

Theorem 3.7 (Properties of Separable Spaces)**.** (a) Every subspace of a separable metric space is separable.

(b) If X is separable and $f : X \to Y$ is continuous and surjective, then Y is separable.

(c) The product $X \times Y$ of two separable spaces is separable.

Proof. (a) Let D be a countable dense subset of X and $Y \subseteq X$. The collection of all sets $Y \cap B_{1/n}(d)$ for $d \in D$, $n \in \mathbb{N}$, is a countable base for Y; choosing one point from each non-empty base element gives a countable dense subset.

(b) Let $D \subseteq X$ be countable and dense. Then $f(D)$ is countable and dense in Y: if $V \subseteq Y$ is open and non-empty, then $f^{-1}(V)$ is open and non-empty in X, so $f^{-1}(V) \cap D \neq \emptyset$, hence $V \cap f(D) \neq \emptyset$.

(c) If $D_X \subseteq X$ and $D_Y \subseteq Y$ are countable dense, then $D_X \times D_Y$ is countable and dense in $X \times Y$ with the product metric. □

Application: Separability is important because it allows us to work with *countable* approximations in spaces of uncountable size. In functional analysis, separable Hilbert spaces have countable orthonormal bases—making Fourier series and spectral theory tractable.

4 Nowhere Dense Sets and Baire's Category Theorem

(Lectures 6–7)

4.1 Nowhere Dense Sets

Definition 4.1 (Nowhere Dense)**.** A subset $N \subseteq X$ is **nowhere dense** if its closure has empty interior:

$$\operatorname{int}(\overline{N}) = \emptyset.$$

Equivalently, every non-empty open set U contains a non-empty open set disjoint from N.

Example 4.1.

- A single point $\{x\}$ in $\mathbb{R}$ is nowhere dense.
- The Cantor set $\mathcal{C} \subset [0,1]$ is nowhere dense (uncountable but has empty interior).
- $\mathbb{Z} \subset \mathbb{R}$ is nowhere dense (every open interval contains reals that are not integers).
- $\mathbb{Q} \subset \mathbb{R}$ is *not* nowhere dense, because $\overline{\mathbb{Q}} = \mathbb{R}$ and $\operatorname{int}(\mathbb{R}) = \mathbb{R} \neq \emptyset$.

4.2 Meager and Non-Meager Sets

Definition 4.2 (First and Second Category)**.**

- A set $M \subseteq X$ is of the **first category** (or **meager**) if it is a countable union of nowhere dense sets.
- A set is of the **second category** (or **non-meager**) if it is not of the first category.

Intuition. Meager sets are "topologically small"—even if they may be dense or uncountable. The Cantor set is uncountable but nowhere dense, hence meager. The rationals are countable but dense, and they are also meager. Baire's category theorem will show that meager sets cannot cover a complete space.

4.3 Baire's Category Theorem

Context and Motivation. Naively, one might expect that any set can be written as a countable union of "thin" (nowhere dense) pieces. The Baire category theorem says this is *impossible* in a complete metric space. This is surprising and powerful: it gives a purely topological tool for proving existence results that cannot be obtained by constructive methods. It is used to prove the existence of continuous functions that are nowhere differentiable, the generic non-existence of well-behaved objects, and various results in functional analysis (Open Mapping, Closed Graph, Uniform Boundedness theorems).

Theorem 4.2 (Baire Category Theorem)**.** A non-empty complete metric space is not a countable union of nowhere dense sets. Equivalently, if

$$X = \bigcup_{n=1}^{\infty} F_n,$$

where each F_n is closed, then at least one F_n has non-empty interior.

Proof. Suppose for contradiction that $X = \bigcup_{n=1}^{\infty} N_n$ where each N_n is nowhere dense. Let $U_n = X \setminus \overline{N_n}$; then U_n is open and dense (since $\overline{N_n}$ has empty interior).

Step 1: Construct a nested sequence of closed balls. Since X is non-empty and $U_1 \neq \emptyset$ (indeed $\overline{N_1} \neq X$), choose $x_1 \in U_1$. Since U_1 is open, there exists $r_1 \in (0, 1/2)$ such that $\overline{B_{r_1}(x_1)} \subseteq U_1$.

Now $B_{r_1}(x_1)$ is non-empty open and U_2 is dense, so $U_2 \cap B_{r_1}(x_1) \neq \emptyset$. Choose $x_2 \in U_2 \cap B_{r_1}(x_1)$. This intersection is open, so we can choose $r_2 \in (0, r_1/2)$ small enough that

$$\overline{B_{r_2}(x_2)} \subseteq U_2 \cap B_{r_1}(x_1).$$

Proceed inductively: choose x_n and r_n such that

$$\overline{B_{r_n}(x_n)} \subseteq U_n \cap B_{r_{n-1}}(x_{n-1}) \quad \text{and} \quad r_n < r_{n-1}/2, \;\; r_n < 1/n.$$

Step 2: Apply Cantor's intersection theorem. We have nested closed balls:

$$\overline{B_{r_1}(x_1)} \supseteq \overline{B_{r_2}(x_2)} \supseteq \cdots, \quad \operatorname{diam}\big(\overline{B_{r_n}(x_n)}\big) \leq 2r_n \to 0.$$

By Cantor's intersection theorem, there is a unique point $x \in \bigcap_n \overline{B_{r_n}(x_n)}$.

Step 3: Contradiction. For any k, $x \in \overline{B_{r_k}(x_k)} \subseteq U_k = X \setminus \overline{N_k}$. So $x \notin \overline{N_k}$, in particular $x \notin N_k$. Since this holds for every k, $x \notin \bigcup_k N_k = X$, contradiction.

Hence X is not a countable union of nowhere dense sets. □

4.4 Applications of Baire Category

Application 1: $\mathbb{R}$ is Uncountable. Suppose $\mathbb{R}$ were countable: $\mathbb{R} = \{x_1, x_2, x_3, \ldots\}$. Each singleton $\{x_n\}$ is closed and nowhere dense. Then $\mathbb{R}$ is a countable union of nowhere dense sets, contradicting Baire's category theorem. (We already proved this directly via Cantor's diagonal, but this is a beautiful alternative proof.)

Application 2: Nowhere Differentiable Continuous Functions. Baire category shows that in $C([0,1])$ (complete under sup norm), the set of nowhere differentiable functions is *comeager*—i.e., "most" continuous functions are nowhere differentiable. This overturned the 19th-century intuition that continuous functions

are "mostly smooth."

Application 3: Uniform Boundedness (Banach–Steinhaus). In functional analysis, the Baire category theorem is used to prove that if a family of bounded linear operators is pointwise bounded, then it is uniformly bounded. This is a key tool in PDE theory and operator theory.

Application 4: $\mathbb{Q}$ is not a G_δ in $\mathbb{R}$. Recall $\mathbb{Q}$ is a countable union of singletons (each nowhere dense), so it is meager. If $\mathbb{Q}$ were a G_δ (countable intersection of open sets), then its complement $\mathbb{R} \setminus \mathbb{Q}$ would be an F_σ, which combined with Baire would give $\mathbb{R}$ as a union of two meager sets, contradiction.

5 Continuous and Uniformly Continuous Functions

(Lectures 8–9)

5.1 Continuity — Review

Definition 5.1 (Continuity at a Point)**.** Let (X, d_X) and (Y, d_Y) be metric spaces. $f : X \to Y$ is **continuous at** $x_0 \in X$ if:

$$\forall \varepsilon > 0,\ \exists \delta > 0 : \quad d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

f is **continuous** (on all of X) if it is continuous at every point of X.

Theorem 5.1 (Sequential Criterion for Continuity)**.** f is continuous at x_0 iff for every sequence $x_n \to x_0$, we have $f(x_n) \to f(x_0)$.

Proof. ($\Rightarrow$) Suppose f is continuous at x_0, and let $x_n \to x_0$. Given $\varepsilon > 0$, choose δ from the continuity definition. Since $x_n \to x_0$, choose N with $d_X(x_n, x_0) < \delta$ for $n \geq N$. Then $d_Y(f(x_n), f(x_0)) < \varepsilon$ for $n \geq N$.

($\Leftarrow$) Suppose the sequential criterion holds but f is not continuous at x_0. Then there is some $\varepsilon > 0$ such that for every $\delta > 0$, there exists x with $d_X(x, x_0) < \delta$ but $d_Y(f(x), f(x_0)) \geq \varepsilon$. Taking $\delta = 1/n$ yields a sequence $x_n \to x_0$ with $f(x_n) \not\to f(x_0)$—contradiction. □

Theorem 5.2 (Open Set Criterion for Continuity)**.** $f : X \to Y$ is continuous (everywhere) iff $f^{-1}(V)$ is open in X for every open set $V \subseteq Y$.

Proof. ($\Rightarrow$) Let $V \subseteq Y$ be open, and pick $x_0 \in f^{-1}(V)$, so $f(x_0) \in V$. Since V is open, $B_\varepsilon(f(x_0)) \subseteq V$ for some $\varepsilon > 0$. By continuity at x_0, choose $\delta > 0$ with $d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon$, i.e., $f(x) \in V$. Thus $B_\delta(x_0) \subseteq f^{-1}(V)$, showing $f^{-1}(V)$ is open.

($\Leftarrow$) Let $x_0 \in X$ and $\varepsilon > 0$. The set $V = B_\varepsilon(f(x_0))$ is open in Y, so $f^{-1}(V)$ is open in X, and it contains x_0. Hence there is $\delta > 0$ with $B_\delta(x_0) \subseteq f^{-1}(V)$, which is exactly the continuity condition. □

5.2 Uniform Continuity

Definition 5.2 (Uniform Continuity)**.** $f : X \to Y$ is **uniformly continuous** if:

$$\forall \varepsilon > 0,\ \exists \delta > 0 : \quad d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon$$

for **all** $x, y \in X$.

Key difference. In continuity at a point, δ may depend on x_0 *and* ε. In *uniform* continuity, δ depends *only on* ε—the same δ works everywhere in the space.

Uniform continuity is a global property: it asserts that f stretches distances uniformly, with a bound independent of the location in X.

Example 5.3 (Continuity Without Uniformity)**.** $f(x) = x^2$ on $\mathbb{R}$. Continuity is clear. But on $\mathbb{R}$, we cannot control the growth: for $x_n = n$ and $y_n = n + 1/n$, we have $|x_n - y_n| = 1/n \to 0$, but $|f(x_n) - f(y_n)| = 2 + 1/n^2 \to 2$. So for $\varepsilon = 1$, no δ works. Hence f is not uniformly continuous on $\mathbb{R}$.

Example 5.4 (x^2 IS Uniformly Continuous on $[0, 1)$.] On a bounded interval, $|x+y| \leq 2$, so $|x^2 - y^2| \leq 2|x - y|$. Take $\delta = \varepsilon/2$. This is uniform.

5.3 Heine–Cantor Theorem

Context. We just saw that x^2 fails to be uniformly continuous on $\mathbb{R}$, but succeeds on $[0, 1]$. What is special about $[0, 1]$? It is *compact*. The Heine–Cantor theorem formalises this: *continuous + compact domain* $\Rightarrow$ *uniformly continuous*. This is one of the deepest and most useful facts in analysis, and it is what makes Riemann integration tractable (every continuous function on $[a, b]$ is Riemann integrable—largely because uniform continuity gives control over the sum of small errors).

Theorem 5.5 (Heine–Cantor Theorem)**.** If X is a compact metric space and $f : X \to Y$ is continuous, then f is uniformly continuous.

Proof. Suppose not. Then there exists $\varepsilon > 0$ such that for every $n \in \mathbb{N}$, we can find $x_n, y_n \in X$ with

$$d_X(x_n, y_n) < \frac{1}{n} \quad \text{but} \quad d_Y(f(x_n), f(y_n)) \geq \varepsilon.$$

Since X is compact and (x_n) is a sequence in X, we can pass to a convergent subsequence: $x_{n_k} \to x$ for some $x \in X$. Since $d_X(x_{n_k}, y_{n_k}) < 1/n_k \to 0$, we also have $y_{n_k} \to x$.

Now f is continuous at x, so $f(x_{n_k}) \to f(x)$ and $f(y_{n_k}) \to f(x)$. Thus $d_Y(f(x_{n_k}), f(y_{n_k})) \to 0$, contradicting the hypothesis that $d_Y(f(x_{n_k}), f(y_{n_k})) \geq \varepsilon > 0$. □

Note on compactness. In metric spaces, compactness (open-cover definition) is equivalent to sequential compactness (every sequence has a convergent subsequence); we will prove this equivalence rigorously in Unit IV.

5.4 Preservation of Cauchy Sequences

Theorem 5.6. Uniformly continuous functions map Cauchy sequences to Cauchy sequences.

Proof. Let $f : X \to Y$ be uniformly continuous and (x_n) Cauchy in X. Given $\varepsilon > 0$, choose $\delta > 0$ from uniform continuity. Since (x_n) is Cauchy, choose N such that $d_X(x_m, x_n) < \delta$ for $m, n \geq N$. Then $d_Y(f(x_m), f(x_n)) < \varepsilon$ for $m, n \geq N$, so $(f(x_n))$ is Cauchy. □

Continuity alone is not enough. Take $f(x) = 1/x$ on $(0, 1]$ and $x_n = 1/n$ (which is Cauchy). Then $f(x_n) = n$, which is *not* Cauchy. This shows the value of uniform continuity: it is precisely the extra hypothesis needed to make Cauchy-ness a *transferable* property.

6 Homeomorphisms

(Lecture 10)

6.1 Definition and Intuition

Definition 6.1 (Homeomorphism)**.** Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is a **homeomorphism** if:

(i) f is bijective;

(ii) f is continuous;

(iii) f^{-1} is continuous.

Two metric spaces are **homeomorphic**, written $X \cong Y$, if there is a homeomorphism between them.

Intuition. Homeomorphism means "topologically equivalent." Two spaces that are homeomorphic have the *same topology*, i.e., their open sets are in exact correspondence. They may look geometrically different (one may be bent, stretched, or shrunk), but they share all topological properties—openness, closedness, compactness, connectedness, etc.

6.2 Examples and Non-Examples

Example 6.1 (Homeomorphic Spaces)**.**

- Any interval (a, b) is homeomorphic to $\mathbb{R}$ via $x \mapsto \tan\big(\pi(x - \frac{a+b}{2})/(b - a)\big)$.
- Any two closed intervals $[a, b]$ and $[c, d]$ are homeomorphic (via an affine map).
- The unit disk in $\mathbb{R}^2$ is homeomorphic to the open unit square (via radial maps).

Example 6.2 (Spaces That are **Not** Homeomorphic)**.**

- $(0, 1)$ and $[0, 1]$: compactness is a topological invariant, and $[0, 1]$ is compact while $(0, 1)$ is not.
- The circle S^1 and the closed interval $[0, 1]$: removing a point from S^1 leaves it connected; removing an interior point from $[0, 1]$ leaves it disconnected.
- $\mathbb{R}$ with the discrete metric and $\mathbb{R}$ with the usual metric: in the discrete metric, singletons are open; in the usual metric, they are not.

6.3 Topological Properties (Invariants)

A property preserved by homeomorphisms is called a **topological invariant**. Some important ones:

- Open sets, closed sets, closure, interior, boundary of subsets.
- Convergence of sequences.
- Continuity of functions.
- Compactness.
- Connectedness.
- Separability.

Warning: Completeness is NOT Topological. Two homeomorphic spaces need not have the same completeness status. Example: $\mathbb{R}$ and $(0,1)$ are homeomorphic (via arctan). $\mathbb{R}$ is complete, but $(0,1)$ is not complete. Completeness is a *metric* property, not a topological one. Similarly, boundedness is not preserved under homeomorphism.

Application: Classification problems. In topology, one of the central goals is to classify spaces up to homeomorphism. For example, the classification of compact surfaces (sphere, torus, Klein bottle, projective plane, etc.) is a landmark result. Every such classification relies on invariants: quantities preserved by homeomorphisms.

7 The Banach Contraction Principle

(Lectures 11–12)

7.1 The Fixed-Point Idea

Definition 7.1 (Contraction)**.** A map $f : X \to X$ on a metric space is a **contraction** (or **strict contraction**) if there exists a constant $k \in [0,1)$ such that

$$d(f(x), f(y)) \leq k\, d(x,y) \quad \text{for all } x, y \in X.$$

The constant k is called the **contraction constant**.

Definition 7.2 (Fixed Point)**.** A point $x^* \in X$ is a **fixed point** of f if $f(x^*) = x^*$.

Intuition. A contraction shrinks distances by a fixed factor $k < 1$. Think of it as a map that pulls everything toward a unique point. If you apply it repeatedly, every trajectory converges to that fixed point—like a heatmap gradually settling to equilibrium.

7.2 Statement and Proof

Context and Applications. The Banach contraction principle is one of the most powerful existence theorems in analysis. It appears in:

- **Differential equations**: Picard–Lindelöf theorem (existence and uniqueness of ODE solutions).
- **Integral equations**: Fredholm and Volterra equations.
- **Optimisation**: convergence of gradient descent and other iterative methods.
- **Fractals**: the existence of attractors in Iterated Function Systems (IFS).
- **Numerical analysis**: Newton's method and iterative solvers are analysed via contraction ideas.

Theorem 7.1 (Banach Contraction Principle)**.** Let (X, d) be a **complete** metric space, and let $f : X \to X$ be a **contraction** with constant $k \in [0, 1)$. Then:

(i) f has a **unique fixed point** $x^* \in X$;

(ii) For any starting point $x_0 \in X$, the iteration $x_{n+1} = f(x_n)$ converges to x^*;

(iii) The rate of convergence is

$$d(x_n, x^*) \leq \frac{k^n}{1-k}\, d(x_1, x_0).$$

Proof. **Step 1: Existence via iteration.** Pick any $x_0 \in X$ and define $x_{n+1} = f(x_n)$.

Step 1a: (x_n) **is Cauchy.** Note:

$$\begin{aligned} d(x_{n+1}, x_n) &= d(f(x_n), f(x_{n-1})) \leq k\, d(x_n, x_{n-1}) \\ &\leq k^2\, d(x_{n-1}, x_{n-2}) \leq \cdots \leq k^n\, d(x_1, x_0). \end{aligned}$$

For $m > n$:

$$d(x_m, x_n) \leq \sum_{j=n}^{m-1} d(x_{j+1}, x_j) \leq \sum_{j=n}^{m-1} k^j\, d(x_1, x_0) \leq \frac{k^n}{1-k}\, d(x_1, x_0) \to 0.$$

Hence (x_n) is Cauchy. Since X is complete, $x_n \to x^*$ for some $x^* \in X$.

Step 1b: x^* **is a fixed point.** Since f is a contraction, it is continuous. So $f(x_n) \to f(x^*)$. But $f(x_n) = x_{n+1} \to x^*$. By uniqueness of limits, $f(x^*) = x^*$.

Step 2: Uniqueness. Suppose $f(x) = x$ and $f(y) = y$. Then

$$d(x, y) = d(f(x), f(y)) \leq k\, d(x, y).$$

Since $k < 1$, this forces $d(x, y) = 0$, so $x = y$.

Step 3: Rate of convergence. From Step 1a, taking $m \to \infty$:

$$d(x_n, x^*) = \lim_{m \to \infty} d(x_n, x_m) \leq \frac{k^n}{1-k}\, d(x_1, x_0).$$

This is our rate bound. □

7.3 Examples and Applications

Example 7.2 (Elementary Example)**.** Let $f(x) = (x+2)/3$ on $\mathbb{R}$ (complete). Then $|f(x) - f(y)| = |x-y|/3$, so f is a contraction with $k = 1/3$. The unique fixed point is $x^* = 1$. Starting from $x_0 = 0$: $x_1 = 2/3$, $x_2 = 8/9$, $x_3 = 26/27$, ..., converging to 1 at rate $(1/3)^n$.

Example 7.3 (Solving $x = \cos x$)**.** Consider $f(x) = \cos x$ on $[0,1]$. By the mean value theorem, $|f(x) - f(y)| = |\sin \xi| \cdot |x-y| \leq (\sin 1)|x-y|$, and $\sin 1 \approx 0.841 < 1$. So f is a contraction with $k \approx 0.841$. Since $[0,1]$ is complete, there is a unique solution to $x = \cos x$ (the Dottie number ≈ 0.739085). This is exactly how numerical solvers find the fixed point.

Application: Picard–Lindelöf (Existence and Uniqueness of ODE Solutions). Consider the initial value problem

$$y'(t) = F(t, y(t)), \quad y(t_0) = y_0.$$

Under a Lipschitz condition on F in y, the associated integral operator

$$(Ty)(t) = y_0 + \int_{t_0}^{t} F(s, y(s))\, ds$$

is a contraction on $C([t_0 - \delta, t_0 + \delta])$ with the sup metric, for small enough δ. By the Banach contraction principle, T has a unique fixed point—which is the solution of the ODE.

Application: Fractal Attractors. In Iterated Function Systems (IFS), a family of contractions $\{f_1, \ldots, f_N\}$ on a complete metric space of compact sets (with the Hausdorff metric) yields a unique "attractor"—a compact set A satisfying $A = \bigcup_i f_i(A)$. This gives a systematic construction of fractals (Cantor set, Sierpinski triangle, Barnsley fern).

Example 7.4 (Showing Completeness is Essential)**.** On the incomplete space $X = (0,1)$ with the usual metric, let $f(x) = x/2$. Then f is a contraction with $k = 1/2$, but has **no fixed point** in X (its fixed point would be $x = 0$, which is outside X). So the Banach theorem genuinely requires completeness.

Example 7.5 (Non-Expansive Map Without Fixed Point)**.** $f(x) = x + 1$ on $\mathbb{R}$. Then $|f(x) - f(y)| = |x-y|$, so f is an isometry (not a strict contraction, $k = 1$). It has no fixed point. This shows $k < 1$ is also essential.

8 Exercises

(Lectures 13–15: Tutorial / Problem-Solving Session)

Part A: Sequences and Cauchy

A1. Prove that if a Cauchy sequence (x_n) in a metric space has a convergent subsequence, then the whole sequence converges.

A2. Let $X = (0, 2)$ with the usual metric. Exhibit an explicit Cauchy sequence in X that does not converge in X.

A3. In a discrete metric space, show that every Cauchy sequence is eventually constant.

A4. Give an example of a metric space X and a sequence (x_n) in X that is Cauchy but not convergent.

A5. Show that if $x_n \to x$ and $y_n \to y$ in a metric space, then $d(x_n, y_n) \to d(x, y)$.

Part B: Completeness

B1. Prove that if F is a closed subset of the complete metric space X, then F is complete.

B2. Show that $(0, 1)$ with the usual metric is incomplete. What is its completion?

B3. Let X be the set of sequences of 0's and 1's with at most finitely many 1's (i.e., the set of eventually-zero binary sequences). Is X complete with respect to the supremum metric $d(x, y) = \sup_n |x_n - y_n|$? Justify using the sequence $(1, 1, \ldots, 1, 0, 0, \ldots)$.

B4. Consider $\mathbb{R}^2 \setminus \{(0, 0)\}$ with the taxicab metric. Show it is not complete by exhibiting a Cauchy sequence (such as $(1/n, 0)$) that does not converge in the set.

Part C: Cantor's Intersection Theorem

C1. Give an example in a **complete** metric space of a decreasing sequence of non-empty closed sets with empty intersection. (Hint: in $\mathbb{R}$, use $[n, \infty)$.)

C2. Verify that $\bigcap_{n=1}^{\infty} [0, 1/n] = \{0\}$ and relate this to Cantor's intersection theorem.

C3. Use Cantor's intersection theorem to show that there is exactly one point in $\bigcap_{n=1}^{\infty} [0, 1/n]$ in $\mathbb{R}$.

Part D: Dense Sets and Separability

D1. Prove that $\mathbb{Q}$ is dense in $\mathbb{R}$.

D2. Prove that $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$ for every n.

D3. Show that ℓ^∞ is not separable. (Hint: for each $A \subseteq \mathbb{N}$, the characteristic function e_A satisfies $\|e_A - e_B\|_\infty = 1$ whenever $A \neq B$.)

D4. Show that the set of all polynomials with rational coefficients is dense in $C([0, 1])$ with the sup metric.

Part E: Baire Category

E1. Use the Baire category theorem to prove that if X is a non-empty complete metric space and $X = \bigcup_{n} F_n$ with F_n closed, then some F_n has non-empty interior.

E2. Show that $\mathbb{Q}$ is of the first category in $\mathbb{R}$.

E3. Show that $\mathbb{R} \setminus \mathbb{Q}$ is not a countable union of closed sets (i.e., is not F_σ in $\mathbb{R}$).

E4. Prove that $\mathbb{Q}$ cannot be written as a countable intersection of open dense subsets of $\mathbb{R}$. (Hint: apply Baire's theorem.)

Part F: Continuity and Uniform Continuity

F1. Show that $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$ but is uniformly continuous on $[0, 1]$.

F2. Show that $f(x) = \sin(x^2)$ is not uniformly continuous on $\mathbb{R}$.

F3. Prove: If $f, g : X \to \mathbb{R}$ are uniformly continuous and bounded, then fg is uniformly continuous.

F4. Show that $f(x) = 1/x$ is not uniformly continuous on $(0, 1]$.

F5. Show directly (without invoking Heine–Cantor) that $f(x) = e^x$ is not uniformly continuous on $\mathbb{R}$.

Part G: Homeomorphisms

G1. Show that $(0, 1)$ and $\mathbb{R}$ are homeomorphic.

G2. Show that $(0, 1]$ and $[0, 1]$ are not homeomorphic.

G3. Show that any two intervals (a, b) and (c, d) are homeomorphic.

G4. Show that the unit circle S^1 is not homeomorphic to a closed interval.

G5. Show that $\mathbb{R}$ with the discrete metric and $\mathbb{R}$ with the usual metric are not homeomorphic.

Part H: Banach Contraction Principle

H1. Let $f(x) = (x + 2)/3$ on $\mathbb{R}$. Verify the Banach contraction principle and find the fixed point.

H2. Show that $f(x) = \cos x$ has a unique fixed point in $[0, 1]$. Estimate it using the Banach iteration starting at $x_0 = 0$, stopping at x_3.

H3. On $(0, 1)$ with the usual metric, show that $f(x) = x/2$ is a contraction but has no fixed point. Which hypothesis of the Banach theorem fails?

H4. Use the Banach contraction principle to prove that the system $x = \sin(y)/3,\ y = \cos(x)/3$ has a unique solution in $\mathbb{R}^2$.

H5. Define $T : C([0, 1]) \to C([0, 1])$ by $(Tf)(x) = \int_0^x f(t)\,dt$. Show that T is not a contraction with respect to the sup metric, but T^2 is.

Part I: Mixed Problems

I1. Give an example of a complete metric space X, a continuous function $f : X \to X$, and a point $x_0 \in X$ such that the orbit (x_n) with $x_{n+1} = f(x_n)$ does *not* converge.

I2. Show that if X is complete and $f : X \to X$ is such that some iterate f^N is a contraction, then f itself has a unique fixed point.

I3. Let $f : [1, \infty) \to [1, \infty)$ be defined by $f(x) = (x + 1/x)/2$. Show that f maps $[1, \infty)$ into itself, is a contraction with constant $k = 1/2$, and has a unique fixed point.

I4. Verify that the set of all Cauchy sequences in a metric space (X, d), under the equivalence relation $x_n \sim y_n$ iff $d(x_n, y_n) \to 0$, forms a complete metric space (this is the standard construction of the completion of X).

Hints to Selected Problems

- **A1.** Use the lemma from Section 1.3: Cauchy plus convergent subsequence implies convergence.
- **B2.** The completion of $(0, 1)$ is $[0, 1]$.
- **B3.** The sequence $(1, 1, \ldots, 1, 0, 0, \ldots)$ is Cauchy but converges (in the sup norm) to the all-ones sequence, which is not finitely supported.
- **B4.** Take $x_n = (1/n, 0)$. This converges (in $\mathbb{R}^2$) to $(0, 0)$, which is not in the set.
- **C1.** In $\mathbb{R}$ with the usual metric, $\{[n, \infty) : n \in \mathbb{N}\}$ is decreasing but has empty intersection.
- **D3.** For each $A \subseteq \mathbb{N}$, the characteristic function e_A has coordinates 1 on A and 0 elsewhere. Distinct e_A, e_B are at distance 1, so any countable dense set can only come within $1/3$ of countably many — contradiction.
- **E4.** Suppose $\mathbb{Q} = \bigcap U_n$ with U_n open and dense. Then $\mathbb{R} \setminus \mathbb{Q} = \bigcup (\mathbb{R} \setminus U_n)$ would be a countable union of closed nowhere dense sets. Combined with $\mathbb{Q}$ being meager, $\mathbb{R}$ would be a countable union of nowhere dense sets — contradicting Baire.
- **F2.** Take $x_n = \sqrt{n\pi/2}$ and $y_n = \sqrt{(n + 1/2)\pi/2}$; then $x_n - y_n \to 0$ but $\sin(x_n^2) - \sin(y_n^2) = \pm 1$.
- **G2.** Compactness is a topological invariant; $[0, 1]$ is compact but $(0, 1]$ is not.
- **H4.** Consider the map $F(x, y) = (\sin y/3, \cos x/3)$ on $\mathbb{R}^2$. Show it is a contraction with constant $1/3$, then apply the Banach theorem.
- **H5.** $\|Tf - Tg\|_\infty \le \|f - g\|_\infty$ (no strict contraction). But $\|T^2 f - T^2 g\|_\infty \le \frac{1}{2}\|f - g\|_\infty$ (strict contraction), so T^2 is a contraction. Then T has a unique fixed point by I2.

— End of Unit III Notes —